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PMC8735764 | 氧化 / 环氧化 | 采集状态 imported_unchecked

标题	Efficient Syntheses of Biobased Terephthalic Acid, p-Toluic Acid, and p-Methylacetophenone via One-Pot Catalytic Aerobic Oxidation of Monoterpene Derived Bio-p-cymene
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出处	ACS Sustainable Chemistry & Engineering 2021
DOI	10.1021/acssuschemeng.1c02605
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摘要（原文节选）

An efficient elevated-pressure catalytic oxidative process (2.5 mol % Co(NO3)2, 2.5 mol % MnBr2, air (30 bar), 125 ° C, acetic acid, 6 h) has been developed to oxidize p-cymene into crystalline white terephthalic acid (TA) in 70% yield. Use of this mixed Co2+/Mn2+ catalytic system is key to obtaining high 70% yields of TA at relatively low reaction temperatures (125 ° C) in short reaction times (6 h), which is likely to be due to the synergistic action of bromine and nitrate radicals in the oxidative process. Recycl ⋯

数据表 · Table 1（8 行，含数值行 8 行）

Optimization of Atmospheric-Pressure Conditions Used for Co(II)/Mn(II) Catalyzed Oxidation of p-Cymene into p-MA, p-Toluic Acid, and Terephthalic Acid

entry	temp (° C)	time (h)	Co source (2.5 mol %)	Mn source (2.5 mol %)	additive (mol %)	yield of p-MA (%)	yield of p-toluic acid (%)	yield of TA (%)
1	120	48	Co(OAc)2	Mn(OAc)2	–	20	2	0
2	120	48	Co(OAc)2	Mn(OAc)2	NaBr (5%)	42	5	0
3a	120	24	CoBr2	Mn(OAc)2	–	60	10	0
4	120	24	Co(NO3)2	Mn(NO3)2	–	12	27	0
5	120	24	CoBr2	Mn(NO3)2	–	26	48	2
6	120	24	Co(NO3)2	MnBr2	–	17	56	7
7	100	24	Co(NO3)2	MnBr2	–	16	54	12
8	100	48	Co(NO3)2	MnBr2	–	9	20	49

脚注（逐字）：aReaction carried out in air.

正文对应句（逐字）：Our attention then turned to identifying catalytic aerobic conditions to oxidize p-cymene into p-toluic acid under more forcing oxidative conditions. A review of the literature revealed that metal nitrate salts had been reported to be more effective than their corresponding acetate salts in mixed me

数据表 · Table 2（6 行，含数值行 6 行）

Optimization of the Elevated-Pressure Oxidation Reaction of p-Cymene to TA

entry	oxidizing gas	T (° C)	isolated yield (%)
1	air or 10% O2 in N2	100	0
2	10% O2 in N2	150	62
3	air	150	70
4	air	175	68
5	air	200	68
6	air	125	70

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